

Homework 4

Due September 23, 2025.

Problem 1. Groups of Lie type.

We discussed in class that all simple finite groups are contained in 4 sets

1. the cyclic groups Z_p of order p (p is a prime number)
2. the alternating groups A_n for $n \geq 5$
3. (not all) groups of Lie type
4. the 26 sporadic groups (we shall not discuss them).

We study here as an introduction the simplest groups of Lie type which are, however, not simple in the technical sense (see (3)).

Consider $GL(2, p)$ which 1) is the set of 2×2 matrices whose entries lie in the Galois field $GF(p)$, meaning that their entries are selected from the integers $0, 1, 2, \dots, p-1$ where p is a prime number, and 2) whose determinant is non zero. Group multiplication is defined as matrix multiplication with entries evaluated mod p . It can be shown that such matrices form a group. For example, for $p = 3$ the entries can be $0, 1, 2 \pmod 3$, or, equivalently, $-1, 0, 1 \pmod 2$, which is easier to work with. A few examples of group elements of $GL(2, 3)$ are as follows $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & -1 \\ -1 & -1 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

- a) Show that the total number of group elements of $GL(2, p)$ is

$$|GL(2, p)| = (p^2 - 1)(p^2 - p). \quad (1)$$

Hint: In how many ways can one choose a, b in $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$?

- b) Show that the group elements with $\det m = +1$ form a subgroup. It is called $SL(2, p)$. What is the order of $SL(2, p)$? *Hint:* What values can $\det m$ take for a matrix m in $GL(2, p)$?
- c) The center of $GL(2, p)$ for $p > 2$ is Z_2 . What are these 2×2 matrices? What is the center of $SL(2, p)$?
- d) By dividing $SL(2, p)$ by its center, one obtains a group called $PSL(2, p)$. What is the order of $PSL(2, 2)$, $PSL(2, 3)$, $PSL(2, 5)$ and $PSL(2, 7)$?
- e) Now consider $GL(2, 2)$. It is defined by all 2×2 matrices with entries 1 or 0. Write down these matrices. What is $SL(2, 2)$? What is $PSL(2, 2)$? Write under each matrix its order. What is the order of $PSL(2, 2)$? Identify $PSL(2, 2)$ with a group we have discussed in class. Write under each 2×2 matrix the permutation to which it is isomorphic.

- f) Show that $PSL(2, 3)$ is isomorphic to A_4 . *Hint:* What is the order of $PSL(2, 3)$? A_4 is not simple. Further $PSL(2, 2)$ is not simple either, so not all $PSL(2, p)$ are simple, but $PSL(2, 5) \simeq A_5$ is simple. In fact, all $PSL(2, p)$ with $p \geq 5$ are simple. (The groups $PSL(2, p)$ with $p \geq 7$ are new groups we have not yet met in class. In Ramond $PSL(2, 7)$ is worked out.)

Problem 2.

Consider the group Q_4 , the quaternion group of order 8. What are its classes? Construct the character table. Which irreps are faithful? Which irreps are real, pseudoreal, or complex? Check your answers by using the Frobenius-Schur criterion.

Problem 3.

Consider the regular representation of Q_4 . Construct the matrices for the 3 group elements I , J , and K . Use the characters of problem 2 to deduce how often the unit irrep and the 2-dimensional irrep appear in the regular representation.